AXIOM-DINOSAUR v4.0

Truthimatics Full-Paint Deterministic Synthesis

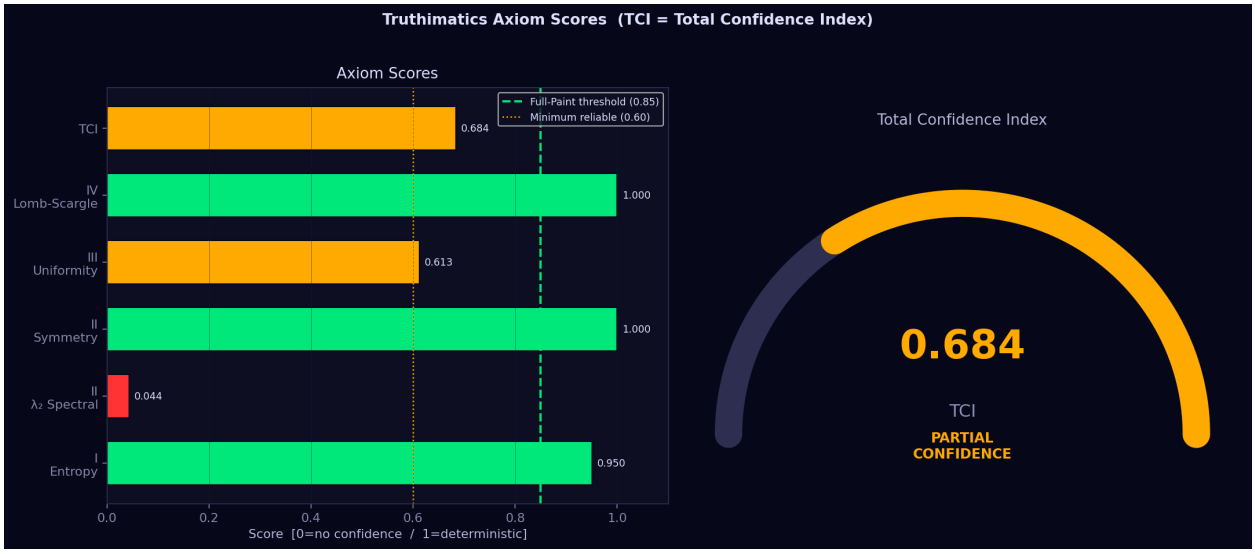
Specimen ID	AXIOM_6B2B3ED4C6
Taxa	Spinosaurus
Clade	Theropod
PBDB Records	44
Mass (regression)	2979.3 kg (Campione & Evans 2012)
Mass (volumetric)	1324.5 kg (Silhouette integral, ρ=1060.0 kg/m³)
Body Volume	1.558 m³
Validity Score	1.0000 / 1.0000
TCI	0.6841 (partial)
Axiom I Entropy	0.9502
Axiom II λ■	0.1763 (symmetry 100%)
Axiom III Uniformity	0.6129
Axiom IV Lomb FAP	1.000000 peak 0.000 Hz
Integrity	56.0% Measured / 44.0% Inferred
Elements	25
CoG (m)	[0.42, 0.969, 0.0]
Generated	2026-04-30 23:52 UTC

Ratio Violations

• skull/femur=2.108 outside [0.30,1.80]

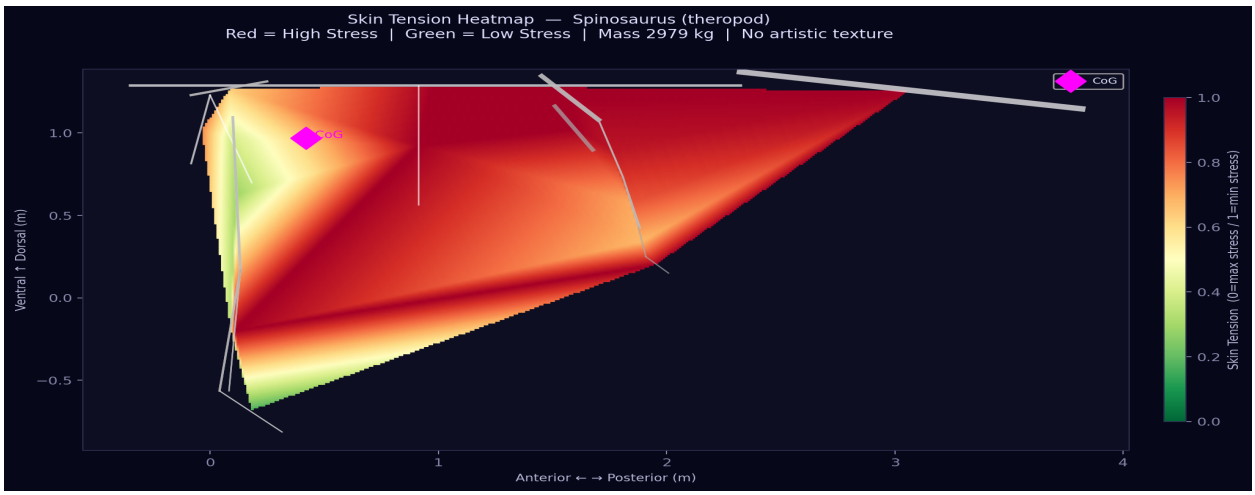
1 Truthimatics Confidence Analysis

TCI = weighted sum: Axiom I (entropy, w=0.25) + Axiom II (spectral, w=0.30) + Axiom III (Biewener uniformity, w=0.30) + Axiom IV (Lomb, w=0.15). TCI ≥ 0.85 authorises Full-Paint output.



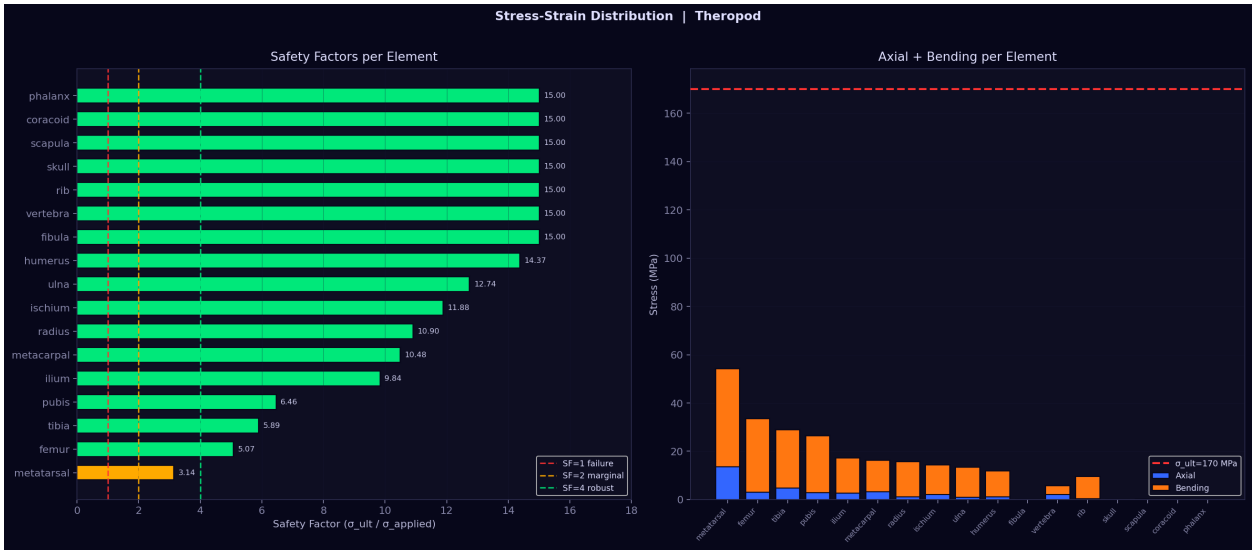
2 Skin Tension Heatmap

Stress field interpolated from per-bone safety factors onto the 2D body silhouette. Red=high stress (low SF), Green=low stress (high SF). No artistic texture applied.



3 Stress-Strain Distribution

Loading: biped (F=Mg). σ_ult=170 MPa (Reilly & Burstein 1975). SF≥4 robust.

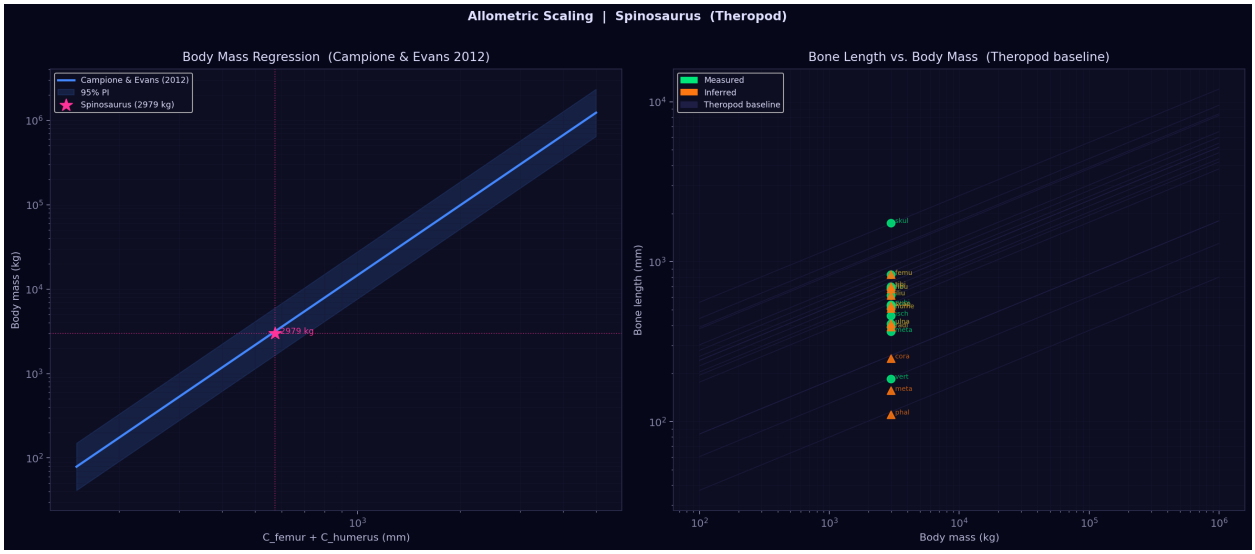


Per-Element Stress Summary

Element	σ_ax MPa	σ_bend MPa	σ_total MPa	SF	Status
coracoid	0.062	0.115	0.177	15.000	OK
femur	3.015	30.538	33.553	5.067	OK
fibula	0.000	0.000	0.000	15.000	OK
humerus	1.126	10.706	11.832	14.368	OK
ilium	2.620	14.664	17.284	9.836	OK
ischium	1.995	12.319	14.314	11.876	OK
metacarpal	3.156	13.058	16.214	10.485	OK
metatarsal	13.492	40.679	54.171	3.138	WARNING
phalanx	0.000	0.000	0.000	15.000	OK
pubis	2.761	23.548	26.310	6.461	OK
radius	1.158	14.436	15.594	10.902	OK
rib	0.279	9.266	9.545	15.000	OK
scapula	0.046	0.157	0.203	15.000	OK
skull	0.000	0.000	0.000	15.000	OK
tibia	4.711	24.145	28.856	5.891	OK
ulna	1.009	12.339	13.348	12.736	OK
vertebra	2.002	3.682	5.684	15.000	OK

4 Allometric Scaling Analysis

Campione & Evans (2012): log■■(M_g)=2.754×log■■(C_f+C_h)–1.097 (R^2=0.97; N=426 non-avian dinosaurs).



5 Physical Properties Analysis

Biomechanical performance estimates derived deterministically from skeletal geometry. Speed: Alexander (1976) locomotion equation. Bite force: Christiansen & Wroe (2007). All values represent physically constrained estimates, not artistic approximations.

Parameter	Value	Notes
Hip Height	1.895 m	Femur + tibia + metatarsal
Estimated Speed	15.2 km/h (4.21 m/s)	Gait: Trot | Fr=0.953
Stride Length	4.284 m	Alexander (1976) / Thulborn (1990)
Bite Force	86392.5 N (8806.6 kgf)	Calibrated theropod regression (Bates & Falkingham 2012)
Skull Length	1750.0 mm	Mid-shaft proxy from skull feature vector
Femoral Robusticity	0.4164	Circumference / length (dimensionless)
Limb Strength Index	3.7e-05	(A_f + A_t) / M^0.67 (Christiansen 1998)
Est. Body Length	7.13 m	Allometric from femur length
Est. Neck Length	1.0 m	Allometric from femur length
Est. Tail Length	2.71 m	Allometric from femur length

6 Complete Bone Manifest

Element	L (mm)	D (mm)	Vol cm³	Mass g	Method	Conf	SF
coracoid	248.8	181.1	3267.96	4079.4	Constraint	0.40	15.00
femur	830.0	110.0	4022.75	5021.6	Measured	0.88	5.07
femur_right	830.0	110.0	4022.75	5021.6	Symmetry	0.84	5.07
fibula	678.0	38.0	276.81	290.3	Measured	0.88	15.00
fibula_right	678.0	38.0	276.81	290.3	Symmetry	0.84	15.00
humerus	510.0	72.0	1059.00	1321.9	Measured	0.88	14.37
humerus_right	510.0	72.0	1059.00	1321.9	Symmetry	0.84	14.37
ilium	615.0	118.0	3430.04	4281.7	Measured	0.88	9.84
ilium_right	615.0	118.0	3430.04	4281.7	Symmetry	0.84	9.84
ischium	460.0	80.0	1179.23	1472.0	Measured	0.88	11.88
metacarpal	156.1	41.7	40.47	33.3	Constraint	0.65	10.48
metatarsal	365.0	52.0	395.33	493.5	Measured	0.88	3.14
phalanx	110.6	37.7	18.99	14.7	Constraint	0.40	15.00
pubis	540.0	68.0	1000.16	1248.5	Measured	0.88	6.46
radius	390.0	42.0	275.56	344.0	Measured	0.88	10.90
radius_right	390.0	42.0	275.56	344.0	Symmetry	0.84	10.90
rib	680.0	33.0	296.62	370.3	Measured	0.88	15.00
scapula	530.0	210.0	9362.13	11686.7	Measured	0.88	15.00
scapula_right	530.0	210.0	9362.13	11686.7	Symmetry	0.84	15.00
skull	1750.0	650.0	296158.92	369695.2	Measured	0.88	15.00
tibia	700.0	88.0	2171.32	2710.5	Measured	0.88	5.89
tibia_right	700.0	88.0	2171.32	2710.5	Symmetry	0.84	5.89
ulna	410.0	45.0	332.56	415.1	Measured	0.88	12.74
ulna_right	410.0	45.0	332.56	415.1	Symmetry	0.84	12.74
vertebra	185.0	135.0	1350.51	1685.8	Measured	0.88	15.00